

Treatment of Asthma & COPD

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 Adapted from: [Drugs to Treat Asthma and COPD](#)

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Learning Objectives (3)

Versions

After completing this brick, you will be able to:

- 1 Justify the use of pharmacological agents in treating asthma and COPD based on their mechanism of action and pharmacokinetic profile.
- 2 Demonstrate an ability to recognize and manage adverse drug reactions for medications used to treat asthma and COPD.
- 3 Design or formulate an evidence-based pharmacotherapeutic approach to treat asthma and COPD.



CASE CONNECTION

You are seeing QS, a 19-year-old college student who is home for a long weekend. His mother called for an urgent appointment because he is having more problems with his asthma. QS says, “I think it’s just the dust in the dorm that is making it worse. And you know, there are a lot of trees and plants on campus, not like what we have here in the city.” His mother adds, “He has gone through two inhalers in 2 weeks.” On exam, QS has scattered wheezes in both lungs. His peak flow is 75% of predicted; it increases to 80% of predicted after two puffs of albuterol inhaler.

Is QS taking the optimal medications for his asthma? Do you need a medication to immediately control the symptoms that is different from long-term therapy? Consider your answer as you read, and we’ll revisit at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Medications Are Used to Treat Asthma and COPD? (LO#3)

Below we will discuss a host of pharmacologic treatments. Because asthma and COPD are both restrictive airway diseases, there is significant overlap in the pharmacologic approaches. Since these are reactive airway diseases mediated by inflammatory processes and parasympathetic tone, the medications we use for treatment are aimed at one, or both, of those processes.

Table 1 provides a brief summary of which medications are utilized for both asthma and COPD, broken down by group or step. Note that because of the primary inflammatory component in asthma, ICS are prioritized first. For COPD, β_2 -agonists and muscarinic antagonists are prioritized first.

Table

	Asthma	COPD
Step 1-3/ Group A	Low dose - ICS As needed SABA +/- Antileukotriene	LABA or LAMA
Step 4/ Group B	Medium dose - ICS + above	LABA + LAMA
Step 5/ Group E	High dose - ICS + above Consider LAMA or LABA Consider biologic therapies	LABA + LAMA + ICS Consider chronic antibiotics Consider roflumilast Consider theophylline

Inhaled Corticosteroids (ICS) (LO#1)

Inhaled corticosteroids (ICS) are considered the most effective first-line treatment for long-term asthma symptom control. For COPD, ICS are for the most severe patients, i.e. they are added on later in treatment. However, if a COPD patient has a blood eosinophil level of 300 or more, then adding an ICS inhaler is particularly useful. The high eosinophil level indicates there is an underlying airway inflammatory disorder that responds well to inhaled corticosteroids.

Some examples of inhaled corticosteroids include beclomethasone, budesonide, and fluticasone. Table 1 below identifies some of the most common ICS used. It is important to note, sometimes, the ICS medications are combined with β_2 -adrenergic agonists, either short- or long-acting, into a single inhaler for patient convenience. The ICS medications suppress the inflammatory process in the bronchial tubes and reduces swelling and narrowing of the airway.

Table

Most common ICS	Most common ICS/ LABA
Fluticasone	Fluticasone/ Salmeterol Fluticasone/ Vilanterol
Budesonide	Budesonide/ Formoterol
Mometasone	Mometasone/ Formoterol
Beclomethasone	

Specifically, ICS inhibit the synthesis of almost all types of cytokines. They also inactivate NF-κB (nuclear factor κ-light-chain-enhancer of activated B cells), which is responsible for the production of TNF-α (tumor necrosis factor α) and various other inflammatory mediators. In addition, although ICS do not directly produce bronchodilation, they enhance the effectiveness of β2-agonists (eg, albuterol) by upregulating β2-receptors on bronchial smooth muscle, thereby increasing responsiveness to adrenergic stimulation.

Adverse Effects (LO#2)

As we discussed in the Foundations of Pharmacology course, systemic corticosteroids have a myriad of adverse effects. Short-term adverse effects examples include hyperglycemia and psychiatric disturbances. Long-term adverse effects are more concerning and include moon face/ buffalo hump, acne, GI bleed risk, and many others. Luckily, ICS avoids many of these adverse effects because the inhaled route allows for a very minor amount of systemic absorption. This means that the most significant adverse effects of ICS use are hoarse voice and oral candidiasis (i.e., thrush). *Candida albicans* is the most common culprit. *To avoid thrush, the patient should rinse their mouth immediately each time after using their inhaler.*



CLINICAL CORRELATION

Corticosteroids are a very broad and effective class of medication. As we have mentioned above, and in our previous conversation of corticosteroids, we want to avoid the myriad of adverse effects, when possible. In cases of severe asthma exacerbation, systemic corticosteroids (like **prednisone**, **methylprednisone**, **dexamethasone**) may be used but we should be cautious about the exposure. Recurrent or chronic use of systemic steroids can lead to significant adverse effects, so we often only escalate to systemic corticosteroids following a severe exacerbation to limit the patient's exposure.

Some early studies suggested that high-dose inhaled corticosteroids can have similar side effects to systemic glucocorticoids. This was especially a concern for pediatric asthma patients as systemic corticosteroids can cause growth suppression/ muscle wasting. More contemporary literature, long-term studies of pediatric patients, showed that ICS, even at high-doses, had no to minimal effect (1-2 mm) on growth of the patients. *This indicates that ICS have very minimal absorption and their benefits on control of asthma far outweigh the risks!*

As you can see in the figure below, ICS are used in every step of therapy for asthma. The only change is as you step up in therapy their frequency, or dose is increased to ensure the patient is receiving an adequate ICS to control their symptoms.

β_2 -adrenergic agonists (LO#1–3)

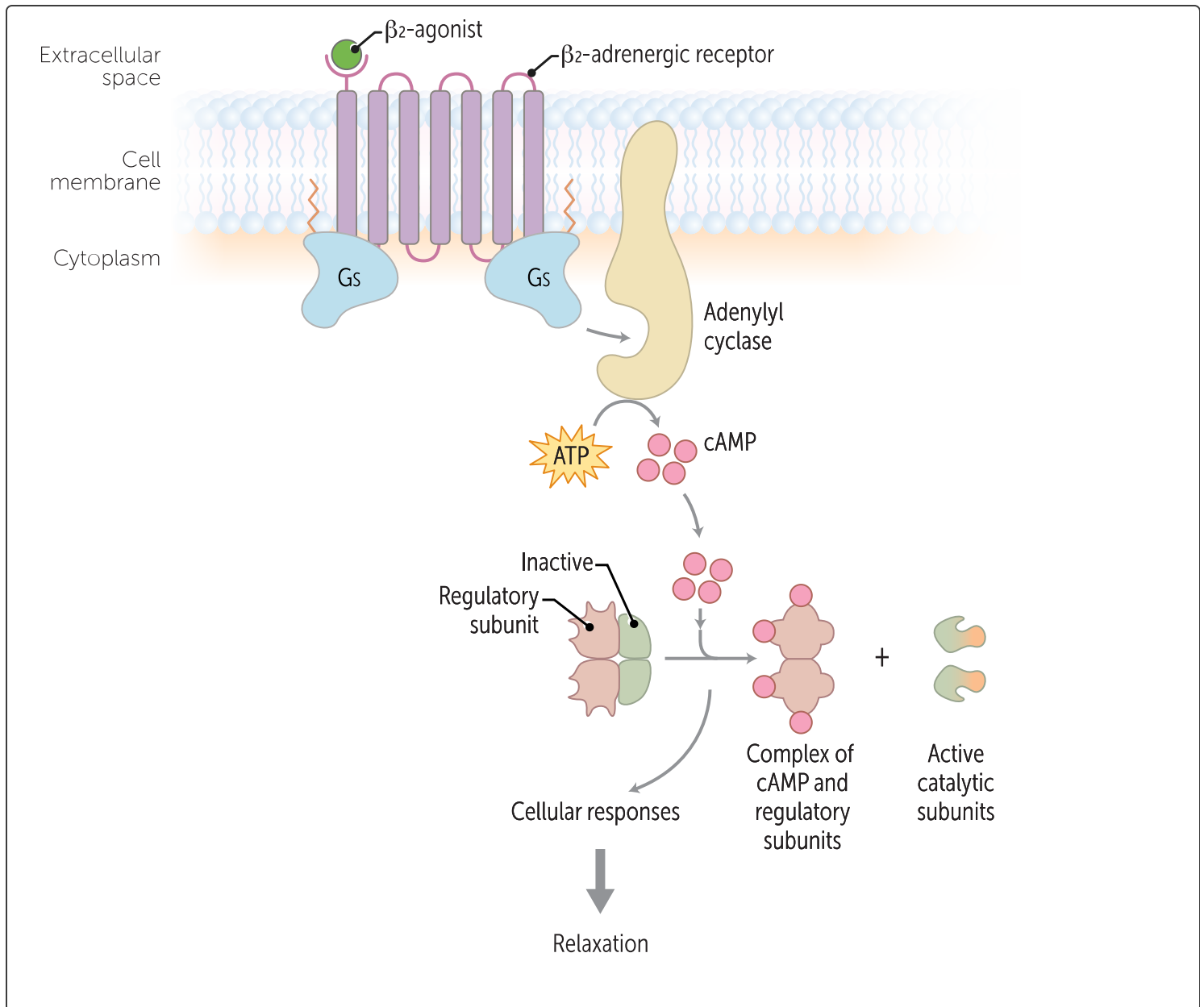
β_2 -adrenergic agonists are another mainstay of therapy for patients with asthma. β_2 -adrenergic agonists are further divided into **short-acting β_2 agonists (SABA)** and **long-acting β_2 agonists (LABA)**, however, their mechanism is the same.

CLINICAL CORRELATION

What makes a SABA short-acting and what makes a LABA long-acting? **SABAs** such as **albuterol** and **levalbuterol** are relatively **hydrophilic**, so they do not remain associated with the cell membrane for long. This allows them to reach the β_2 -receptor quickly, producing **rapid bronchodilation within minutes**, but their effect is brief. In contrast, **LABAs** such as **salmeterol** and **formoterol** have **lipophilic side chains** that partition into the plasma membrane and remain near the receptor, prolonging receptor interaction and extending their duration of action to **12 hours or more**. This structural difference explains why SABAs are used as **rescue inhalers**, whereas LABAs are used for **maintenance therapy**.

The β_2 receptor is a G protein–coupled receptor that activates adenylyl cyclase on stimulation. When activated, the adenylyl cyclase produces cyclic adenosine monophosphate (cAMP), which leads to smooth muscle relaxation

(Figure 1). β_2 receptors are found on smooth muscles in the lung, which leads to the relaxation and dilation of the bronchial passages. This dilation, in turn, leads to easier movement of air for the patient and reduces symptoms caused by airway restrictions.



QUIZ

Tap image for quiz

Figure 1

Adverse Effects (LO#2)

Albuterol is the most well-known example of a SABA. It is indicated for acute asthma exacerbations but is not used as a maintenance therapy to prevent future asthma attacks due to the short duration of action. Albuterol is generally well tolerated with few significant adverse effects. Most commonly, patients may experience mild adverse reactions such as rapid heart rate (i.e., tachycardia), skeletal muscle tremors, and occasionally hypokalemia.



CLINICAL CORRELATION

Recall that β_2 agonists can activate the Na^+/K^+ -ATPase pump to drive Na out of the cell and K into the cell. This essentially shifts potassium intracellularly. This can cause hypokalemia, but this mechanism is also used for the treatment of patients with hyperkalemia where albuterol is one of the primary therapies.

Examples of LABAs include **salmeterol** and **formoterol**. LABAs are used as adjunct therapy alongside glucocorticoids for asthma patients with moderate to severe symptoms that are inadequately controlled with glucocorticoids alone. **They should not be prescribed as monotherapy for a patient with asthma.** They are administered as a daily controller inhaler for asthma exacerbation prophylaxis. Adverse reactions are generally secondary to LABAs' vasodilatory properties and include dizziness, headaches, and rapid heart rate.

to severe asthma symptoms.

LABAs are used as adjunct therapy with glucocorticoids for moderate

Leukotriene Receptor Antagonists (LO#1–3)

The **anti-leukotrienes** make up a class of drug that alleviates asthma symptoms by inhibiting the actions of leukotrienes—a group of potent inflammatory mediators. Members of this class include **montelukast**, **zafirlukast**, and **zileuton**. Montelukast and zafirlukast work by blocking the cysteinyl leukotriene receptor on target cell surfaces, which is activated by leukotrienes. Zileuton inhibits leukotriene production by inhibiting circulating lipoxygenase.

Anti-leukotrienes, primarily montelukast, are common add on/ adjunct therapy. There are two common patients that benefit most from antileukotrienes: 1) Patients whose symptoms are not completely controlled with their ICS and have concerns about increasing their ICS dose, and 2) patients with specific triggers of exercise, allergen induced, and allergic rhinitis.

Adverse Effects (LO#2)

Montelukast carries and FDA black box warning for neuropsychiatric effects, primarily related to mood changes/ disorders. This was primarily observed in pediatric patients, though more recent data has shown that it is much less common than initially thought and relatively rare. Zileuton is specifically associated with hepatotoxicity.

effects, including mood changes.

Montelukast has an FDA black box warning for neuropsychiatric

Muscarinic Antagonists (LO#1)

Muscarinic antagonists work by competitively blocking muscarinic receptors (M3-receptor) on the airway smooth muscles and preventing bronchoconstriction. They are divided into long-acting muscarinic antagonists (LAMA) and short-acting muscarinic antagonists (SAMA). A common short-acting muscarinic antagonist agent is **ipratropium**, and a long-acting muscarinic antagonist is **tiotropium**.

Ipratropium is more commonly used to treat COPD and not usually used as a first-line bronchodilator for chronic asthma. Currently, it is usually used in emergency departments as adjunct therapy for acute asthma exacerbation. Adverse effects of ipratropium include headache, dizziness, nausea, and dry mouth. These are essentially the same side effects seen with anticholinergics (Figure 3).

Adverse Effects (LO#2)



a Flask' **Mad** as a Hatter

Blind as a Bat' **Hot** as a Hare' **Dry** as a Bone' **Red** as a Beet' **Full** as

What are the adverse effects of muscarinic antagonists?

LAMA medications, like **tiotropium**, also have more of a role in COPD than asthma. However, patients who have severe asthma, Step 5, may benefit from adding a LAMA to their medication regimen.

Phosphodiesterase Inhibitors (Theophylline and Roflumilast) (LO#1)

Both **theophylline** and **roflumilast** are **phosphodiesterase** inhibitors, though slightly different. Theophylline is a close relative to **caffeine**. It works by inhibiting phosphodiesterase (PDE) enzymes, primarily PDE3 and PDE4, which prevents the breakdown of cyclic AMP and GMP. This leads to increased intracellular cAMP, resulting in bronchodilation of airway smooth muscle and anti-inflammatory effects through reduced pro-inflammatory mediator release and enhanced **interleukin-10** activity. Both mechanisms promote airway smooth muscle relaxation and bronchodilation (**Figure 3**). Roflumilast, on the other hand, is selective for PDE4. Thereby increasing cAMP levels resulting in bronchodilation.

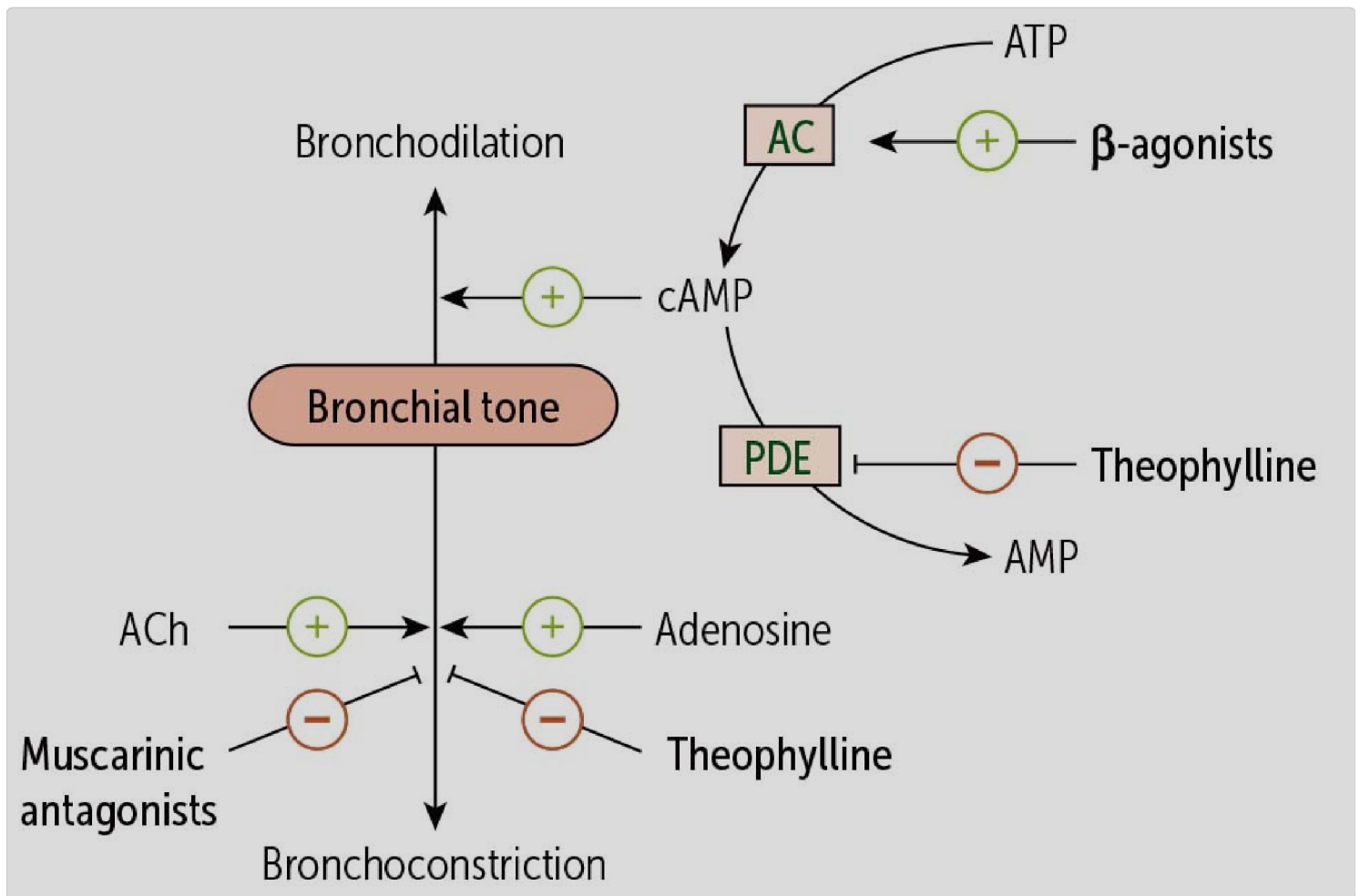


Figure 3

Adverse Effects (LO#2)

Theophylline can be used to enhance bronchodilation if other agents have failed, but it is not considered first-line. There are multiple reasons that theophylline has fallen out of favor, despite being one of the very first medications used for asthma. First, its side effect profile is concerning. Adverse effects include nervousness, tremor, and tachycardia. In addition, theophylline is a substrate of CYP1A2 and CYP3A4, so there are significant drug-drug interactions that may occur. Second, it is not effective as other newer medications that are available. Lastly, it has a narrow therapeutic index that requires serum monitoring to prevent toxic levels. So, it is typically reserved for acute asthma exacerbations where patients have failed to respond to other first-line therapies. It can also be used for very severe COPD patients who have failed to respond to other therapies as well.

Roflumilast is considered an add-on therapy in select patients with severe COPD, particularly those with chronic bronchitis and frequent exacerbations. It is not approved for asthma treatment. Its use is limited for several reasons. First, it is less effective for bronchodilation compared to standard inhaled therapies. Second, it carries a notable side effect profile, including weight loss, gastrointestinal upset, and psychiatric effects such as anxiety or depression. Lastly, because it primarily reduces exacerbations rather than providing immediate symptom relief, its role is more specialized. As a result, roflumilast is generally reserved for patients with severe COPD who remain symptomatic despite maximal inhaled therapy.

therapeutic index requiring serum monitoring.
interactions, less effective than newer medications, and a narrow
Theophylline has a concerning side effect profile, significant drug

Cromolyn Sodium (LO#1,2)

Asthma symptoms are largely driven by mast cell degranulation, which initiates an inflammatory cascade and the release of mediators such as histamine. Mast cell stabilizers, like **cromolyn sodium**, act by preventing mast cell degranulation and thereby limiting the release of these mediators. Historically, cromolyn was an important early therapy for asthma, and it can still be used prophylactically before exposure to a known trigger or as long-term maintenance therapy. It is well tolerated, with minimal side effects aside from occasional throat irritation and coughing.

However, cromolyn is rarely used in asthma treatment today because it has been outperformed by ICS, which provide greater improvements in lung function, stronger and more consistent anti-inflammatory effects, and better overall asthma control. Additionally, cromolyn's onset of action is delayed, requiring 2–4 weeks of consistent use before full benefits are seen, and its effects are less predictable than modern therapies. Cromolyn is not recommended for COPD, since its mechanism targets inflammatory pathways rather than the airway remodeling and obstruction that dominate in COPD.

anti-inflammatory effects, and asthma control.

Cromolyn is outperformed by ICS, which offer better lung function,

Biologic Therapies (LO#1)

Biologic therapies are targeted monoclonal antibodies used for patients with moderate to severe asthma who remain uncontrolled despite optimized inhaled therapy. Unlike broad anti-inflammatory medications, these biologics act on specific immune pathways. For example, mepolizumab and benralizumab block interleukin-5 (IL-5) or its receptor, reducing eosinophil survival and activity. Dupilumab targets the IL-4 receptor alpha, blocking IL-4 and IL-13 signaling, which decreases type 2 inflammation. Omalizumab binds circulating IgE, preventing IgE from interacting with receptors on mast cells and basophils, thereby reducing allergic inflammation.

In terms of place in therapy, biologics are not first-line agents. They are considered as add-on treatment in patients with severe eosinophilic asthma, allergic asthma, or oral corticosteroid-dependent asthma, particularly if blood eosinophil levels are elevated or IgE levels are high and allergen sensitivity is documented. These drugs help reduce exacerbations, improve lung function, and allow tapering of systemic corticosteroids. Selection

depends on the asthma phenotype: omalizumab for allergic asthma; mepolizumab, benralizumab, or dupilumab for eosinophilic asthma; and dupilumab also for patients with overlapping conditions like atopic dermatitis or chronic rhinosinusitis with nasal polyps. A summary of the biologic therapies for asthma is in the [table](#) below.

Table

Name	Molecular Target	Selection Criteria	Special Indications
Omalizumab	IgE	Elevated IgE, Improved efficacy if Eosinophils or elevated FeNO	Childhood asthma, polyps
Dupilumab	IL-4 Receptor (IL-4, IL-13)	Peripheral Eosinophils, Elevated FeNO	Atopic Dermatitis, Nasal Polyposis
Mepolizumab	IL-5	Peripheral Eosinophils	EGPA
Benralizumab	IL-5 Receptor	Peripheral Eosinophils	Best tolerated (dosed every 2 months)

Adverse Effects (LO#2)

The adverse effect profile of biologics is generally milder compared to long-term corticosteroid use, but each drug carries unique considerations. Common effects include injection-site reactions and headache. Mepolizumab and benralizumab may cause hypersensitivity reactions and increase risk of certain infections (e.g., herpes zoster, helminths). Dupilumab is associated with conjunctivitis, eosinophilia, and occasional joint symptoms. Omalizumab carries a boxed warning for anaphylaxis, which can occur even after long-term use. Despite these risks, biologics have transformed the care

of severe asthma by offering targeted, phenotype-driven therapy with fewer systemic side effects than oral corticosteroids.

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to QS, is he taking the optimal medications for his asthma?

You say, “Whatever the initial trigger was, you are in the midst of an asthma flare-up. The albuterol inhaler is fine for a sudden episode, but it is clearly not enough, and it is not doing anything to fix the inflammation that is beneath the surface.” You propose to treat him with a short course of prednisone, to quickly make an impact, and to give him a steroid inhaler. “You can continue to also use your albuterol. But the main thing that will fix this is the steroid medicine. Once we get this flare-up under control, we should keep you on the steroid inhaler for a period of time, perhaps another month. We may need to keep you on this long-term. It is much more effective than the albuterol for long-term control.” With his prescriptions, you give QS a written summary of the treatment plan for him to keep and to share with student health at his college.

Summary

- Asthma severity is determined by symptom frequency, nighttime awakenings, inhaler use, and daily life interference.
- Inhaled corticosteroids (ICS) are the most effective first-line treatment for long-term asthma control, with minimal systemic absorption.
- β 2-adrenergic agonists, divided into short-acting (SABA) and long-acting (LABA), cause bronchodilation by stimulating β 2 receptors on bronchial smooth muscles.

- Muscarinic antagonists block muscarinic receptors on airway smooth muscles, preventing bronchoconstriction, and are used in severe asthma cases.
- Theophylline, a phosphodiesterase inhibitor, is not first-line due to side effects, drug interactions, and narrow therapeutic index.
- Biologic therapies target specific immune pathways and are used for severe asthma uncontrolled by inhaled therapies.

Review Questions

1. Five weeks after her 13-year-old daughter began taking a new asthma drug, a worried mother brought her to her primary care doctor because of some “new growth” in her mouth. On examination, there appears to be a white plaque coating her tongue. The plaque was easily scraped off with a tongue depressor. Which of the following is most likely responsible for this patient’s presentation?

- A. Albuterol
- B. Budesonide
- C. Ipratropium
- D. Miconazole
- E. Montelukast

Explanation (requires correct answer)

2. A mother brought her 12-year-old boy to the physician because she noticed that every time he plays soccer in the cold, he becomes short of breath. She explains that his symptoms are so severe that sometimes he has to leave the game, gasping for air. A diagnosis of exercise-induced asthma was made, and the boy was given a short-acting β_2 agonist. What is the main mechanism of action of this medication?

- A. Adenylyl cyclase inhibition
- B. Decreased leukotriene production
- C. Increased cAMP production
- D. Inactivation of NF- κ B
- E. Phosphodiesterase inhibition

Explanation (requires correct answer)

3. Which of the following can be used specifically for atopic asthma due to its IgE binding effect?

- A. Albuterol
- B. Corticosteroids
- C. Ipratropium bromide
- D. Montelukast
- E. Omalizumab

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Justify the use of pharmacological agents in treating asthma and COPD based on their mechanism of action and pharmacokinetic profile.
- 02 Demonstrate an ability to recognize and manage adverse drug reactions for medications used to treat asthma and COPD.
- 03 Design or formulate an evidence-based pharmacotherapeutic approach to treat asthma and COPD.

Objective Confidence: 0/3